



## PAPER

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# Bidirectional Mamba state-space model for anomalous diffusion

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## Abstract

Characterizing anomalous diffusion (AnDi) is crucial in order to understand the evolution of complex stochastic systems, from molecular interactions to cellular dynamics. Here, we evaluate the performances regarding such a task of Bi-Mamba, a novel state-space deep-learning architecture articulated around a bidirectional scanning mechanism. Our implementation is tested on the AnDi-2 challenge datasets. Designed for regression tasks, the Bi-Mamba architecture infers efficiently the effective diffusion coefficient and anomalous exponent from single, short trajectories. As such, our results indicate the potential practical use of the Bi-Mamba architecture towards the characterization of AnDi.

## 1. Introduction

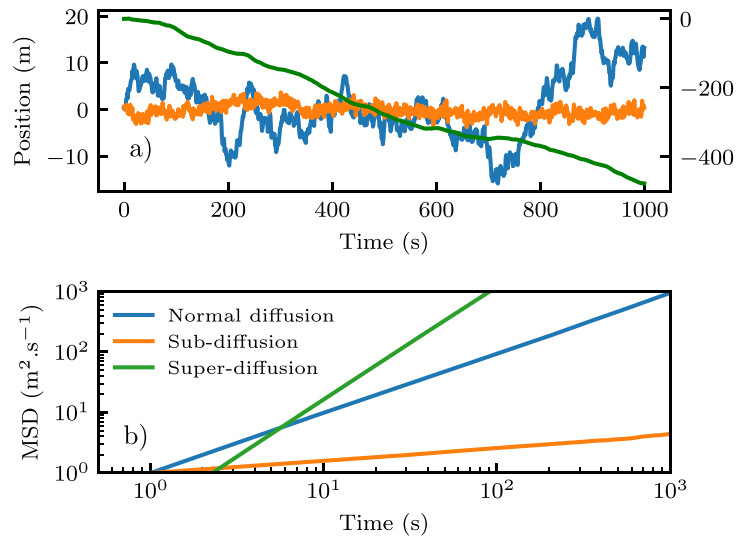
Anomalous diffusion (AnDi) [1] is ubiquitous in complex stochastic processes, and central in particular to the cell machinery, as classically shown by microrheology [2]. Its understanding provides insights on the transport of microparticles in heterogeneous media, such as molecules in biological cells [3, 4]. Methods such as microscopy combined with single-particle tracking allow for the detailed spatiotemporal analysis of single events [5, 6]. Over the last few decades, single-particle imaging methods have been steadily upgrading, increasing the amount of experimental data available on molecular interactions, cellular dynamics, microparticle behaviors [7, 8], and nanoparticles [9] in various complex media. Other approaches, such as dynamic light scattering [10, 11] and differential dynamic microscopy [12] have provided valuable insights on dense particle suspensions.

However, in most biological cases, trajectories are typically scarce, short and noisy. As such, it is often hard, if not impossible, to infer meaningful information using traditional tracking methods. Deep-learning methods for advanced microscopy have thus emerged as a promising change of paradigm [13]. As such, the work we present here results from the AnDi challenge [14], and in particular from our participation to its second edition [15]. We took a step aside from the standard methods used by the other participants in order to improve the collective knowledge on advanced artificial-intelligence methods for complex particle tracking. Our results indicate that the novel Bi-Mamba model infers efficiently the effective diffusion coefficient and anomalous exponent from single, short and noisy trajectories.

In a typical two-dimensional anomalous-diffusion process, the mean-squared displacement (MSD) is determined over time  $t$  by:

$$\text{MSD}(t) = 4Kt^\alpha, \quad (1)$$

where  $K$  is the effective diffusion coefficient and  $\alpha$  is the AnDi exponent. For Brownian particles,  $\alpha = 1$ . Besides, a particle is in a sub-diffusive regime when  $0 < \alpha < 1$ , and in a super-diffusive regime when  $1 < \alpha < 2$ . In figure 1, typical samples of these different diffusion types are shown with their respective trajectories and MSDs. For example, in complex cell environments a particle can exhibit AnDi due to:



**Figure 1.** (a) Typical trajectories numerically simulated for a Brownian particle ( $\alpha = 1$ ), a sub-diffusive particle with  $\alpha = 0.2$ , and a super-diffusive particle with  $\alpha = 1.98$ . (b) Corresponding mean-squared displacements of the trajectories (see equation (1)).

- *Obstacles and crowding*: in the presence of obstacles, diffusion can be hindered, leading to a sub-diffusive behavior. Moderate concentrations of obstacles can cause AnDi over short distances, and the effect becomes more pronounced as the obstacle concentration approaches the percolation threshold [16];
- *Binding and trapping*: temporary binding of particles to fixed sites [17] leads to a sub-diffusive behavior;
- *Lipid rafts and membrane heterogeneity*: interactions of particles with lipid rafts in cell membranes have been shown to lead to complex diffusive behaviors [18].

The first AnDi Challenge took place in 2020 and aimed at assessing the performances of various methods in quantifying AnDi [14]. The main focus was dedicated to the inference of  $K$  and  $\alpha$ , for various simulations of bio-mimetic cases. The second AnDi challenge took place in 2024 and aimed at evaluating the performances of various methods for detecting and quantifying changes in single-particle motion [15]. As such, the focus was not only on the inference of the effective diffusion coefficient and exponent, but also on trajectory segmentation, with the trajectory statistics being described by a maximum of two  $K$  and two  $\alpha$  values. One typical goal is for example to measure at what rate a particle binds and unbinds to/from a given cellular site. The AnDi datasets involve five phenomenological models:

- *Single-state*: particles diffusing according to a single diffusion state, as observed for some lipids in the plasma membrane [19–21];
- *Multi-state*: particles diffusing according to two diffusion states, and undergoing transient changes of  $K$  and  $\alpha$ , as observed for proteins due to allosteric changes or ligand binding [22–25];
- *Dimerization*: particles diffusing according to two diffusion states, and undergoing transient changes of  $K$  and  $\alpha$ , induced by encountering other diffusing particles, as observed in protein dimerization and protein-protein interactions [26, 27];
- *Transient confinement*: particles undergoing transient diffusion changes when entering or leaving given areas, as observed in the confinement induced by clathrin-coated pits on cell membranes [28];
- *Quenched trap*: two-state model of diffusion representing proteins being transiently immobilized at specific locations, induced by binding to immobile structures, as observed in cytoskeleton-induced molecular pinning [29, 30].

## 2. Bidirectional state-space model

Deep-learning models for inference from time series are generally based on recurrent neural networks (RNNs) or convolutional neural networks (CNNs). RNNs process the data sequence by updating a hidden state after each element. However, as the gradient also needs to go through each element one by one, long-term information may be lost. CNNs use convolutional kernels to give more weight to local information, as they do not need to process the sequence one by one. CNNs benefit from parallel computing and faster training speed but also limit the retrieval of long-term global information.

**Algorithm 1.** Bi-Mamba.

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```

Input:  $x \in \mathbb{R}^{(B,L,D)}$ 
Output:  $y \in \mathbb{R}^{(B,L,D)}$ 
1  $x \leftarrow \text{Norm}(x)$ ;
2 for  $i \in \{\text{forward}, \text{backward}\}$  do
3   if  $i = \text{forward}$  then
4      $x_i \leftarrow x$ ;
5   else
6      $x_i \leftarrow \text{Flip}(x)$ ;
7   end
8    $x_i \leftarrow \text{MambaBlock}(x_i)$ ;
9   if  $i = \text{forward}$  then
10     $x_i \leftarrow \text{Flip}(x_i)$ ;
11  end
12   $x_i \leftarrow \text{Norm}(x_i + x)$ ;
13 end
14  $y_M = x_{\text{forward}} + x_{\text{backward}}$ ;
15  $y \leftarrow \text{FeedForward}(y_M)$ ;
16  $y \leftarrow \text{Norm}(y + y_M)$ ;
17 return  $y$ ;

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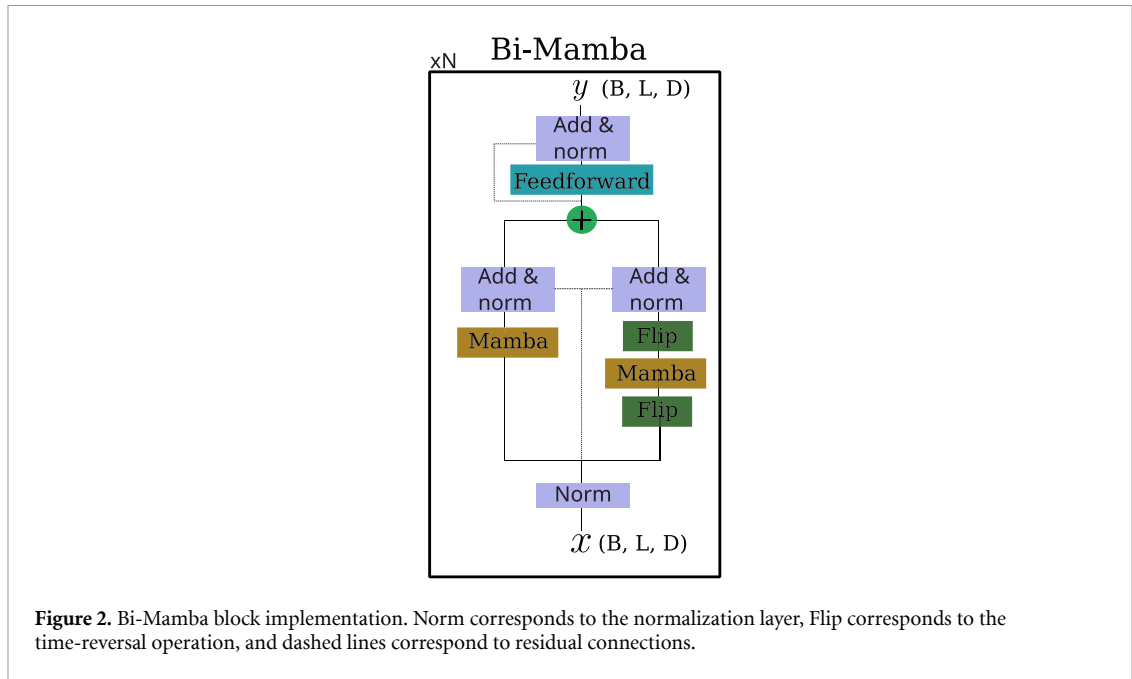
The Mamba architecture [31] is a state-space model designed to overcome the limitations of traditional deep-learning models like RNNs and CNNs by handling long-term dependencies with a high training efficiency. The key innovations of the Mamba architecture include a selection mechanism and hardware optimization for GPU usage. The Mamba architecture is essentially a reformulation of RNNs and CNNs as selective state-space models. The novelties brought by this architecture consist in:

- *A selection mechanism:* the selection mechanism in the Mamba model enables it to dynamically focus on relevant information and ignore irrelevant data. This mechanism is akin to the attention mechanism used in Transformer models. It allows the model to adjust its focus based on the input data, enhancing its ability to capture important features and dependencies;
- *Hardware awareness:* the Mamba architecture is optimized for GPU usage, which allows it to scale linearly with sequence length. This optimization ensures that the model can handle long sequences efficiently and benefit from parallel computing, resulting in faster training.

The Mamba architecture was originally developed for use in large language models (LLMs). However, recent studies in the literature have demonstrated its capability to perform segmentation on 3D medical images, showcasing its versatility and robustness in handling complex data structures. Therefore, the present study aims at determining whether a Mamba-based model can be effectively applied to complex time-series datasets, such as anomalous Brownian trajectories. The approach we chose was to evaluate the performances of Bi-Mamba, i.e. the novel state-space deep-learning architecture (Mamba) articulated with a bidirectional scan mechanism. Indeed, to the best of our knowledge, the Mamba architecture has not been used so far in the case of physical studies, and AnDi data in particular.

### 2.1. The Bi-Mamba block

Originally, the Mamba architecture has been designed to process 1D data sequence, in one direction. Considering the complex patterns that can occur in anomalous Brownian trajectories, we have implemented a bidirectional version of the Mamba block. As described in the algorithm 1, the block is made as an encoder and takes as an input  $x \in \mathbb{R}^{(B,L,D)}$  with  $B$  the number of trajectories to be analyzed in parallel,  $L$  the length of the trajectories and  $D$  the number of features extracted from the trajectories. The output of the Bi-Mamba block is  $y \in \mathbb{R}^{(B,L,D)}$ . The Bi-Mamba block is composed of two Mamba blocks, one for the forward pass and the other for the backward pass. The forward Mamba block processes the input data in the forward direction, while the backward Mamba block processes the input data in the reverse direction. We denote the time-reversal operation as Flip, and the variables of each side as  $x_{\text{forward}}$  and  $x_{\text{backward}}$ , respectively. The outputs of the two Mamba blocks are then added and passed to a feedforward layer for further processing. To reduce the risk of vanishing gradients, residual connection between the input and the output of each step is applied [32].




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**Algorithm 2.** Our Model—MambaBrown.

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**Input:**  $x \in \mathbb{R}^{(B,L,D)}$

**Output:**  $(y_S, y_K, y_\alpha)$

```

1  $y_S \leftarrow \text{BiMambaBlock}(x)$ ;
2  $y_S \leftarrow \text{Dense}(y_S)$ ; //  $(B, L, 2)$ 
3  $y_K \leftarrow \text{BiMambaBlock}(x, y_S)$ ;
4  $y_K \leftarrow \text{Dense}(y_K)$ ; //  $(B, 2)$ 
5  $y_K \leftarrow \text{ReLU}(y_K)$ ;
6  $y_\alpha \leftarrow \text{BiMambaBlock}(x, y_S)$ ;
7  $y_\alpha \leftarrow \text{Dense}(y_\alpha)$ ; //  $(B, 2)$ 
8  $y_\alpha \leftarrow \text{Sigmoid}(y_\alpha)$ ;
9 return  $(y_S, y_K, y_\alpha)$ ;

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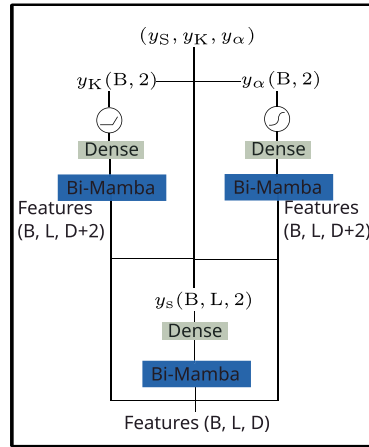
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## 2.2. Our model—MambaBrown

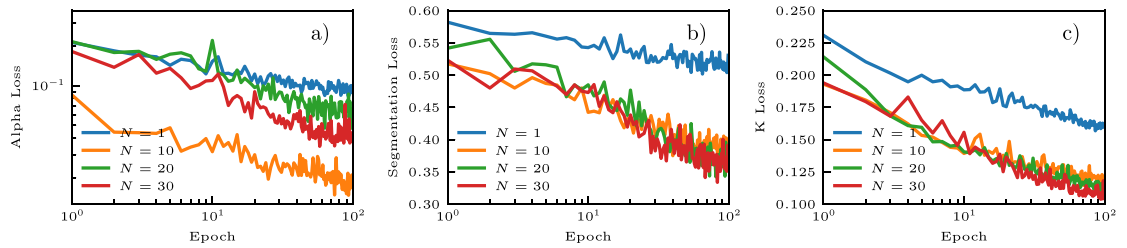
To achieve the different tasks of the AnDi-2 challenge, we have implemented a model using the Bi-Mamba block, as shown in figure 2. The input data is first processed by a Bi-Mamba block for the segmentation of the diffusion state of the trajectory. As the trajectories are simulated using two different diffusion states, the segmentation block output is then processed by a dense layer to reduce the output dimensionality to  $(B, L, 2)$ , with each dimension corresponding to the probability of the trajectory to be in one of the two states. Then, the original features and the segmentation result are passed into two separate blocks, one for the  $K$  regression and the other for the  $\alpha$  regression. Therefore, each task is handled by a dedicated block, allowing each block to specialize in a single function. Each of the outputs of the regression blocks is then processed by a dense layer to reduce the dimensionality of the output to  $(B, 2)$ , with each dimension corresponding to the predicted value of  $K$  and  $\alpha$  respectively. The output of the  $\alpha$ -regression block is passed to a sigmoid activation function and multiplied by 2 to ensure that the output is between 0 and 2. The output for the  $K$ -regression block is passed to a ReLu activation function to ensure that the output is positive.

## 3. Training and results

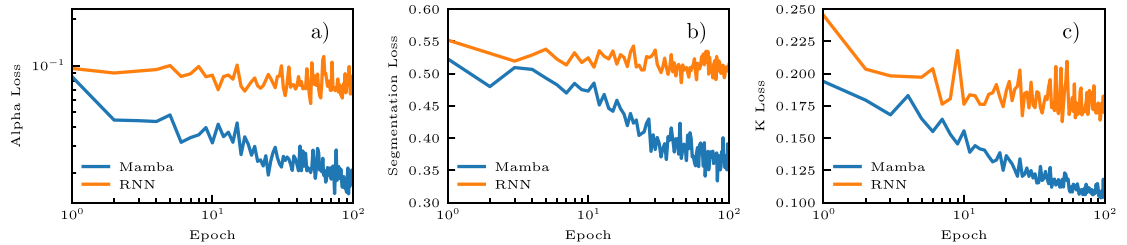
The data ensemble consists of sets of two-dimensional trajectories with coordinates  $r(t) = [r_x(t), r_y(t)]$ , with a maximum length of 200 time steps. The features computed using given trajectories are: the displacements for the first lag time  $\Delta t = 1$ ,  $\Delta r_i(\Delta t = 1) = r_i(t + \Delta t) - r_i(t)$ , with  $i \in \{x, y\}$ , the one-dimensional MSDs  $\langle \Delta r_i^2 \rangle(\Delta t)$ , the displacement angle between two consecutive displacements [33], and the total displacement from the origin  $d = \sqrt{x(t)^2 + y(t)^2}$ . Trajectories smaller than 200 data points are zero-padded. The loss is then calculated by summing the losses of the three blocks and back-propagated using the Adam optimizer. The  $\alpha$  loss is evaluated using mean absolute error (MAE), the  $K$  loss is evaluated using mean-squared



**Figure 3.** MambaBrown model implementation. Dense corresponds to the dense layer reducing the dimensionality of the Bi-Mamba block output. Circles correspond to the activation functions ReLu and Sigmoid, on the left and right respectively.



**Figure 4.** Loss values calculated on a test dataset consisting of 200 000 trajectories, as functions of the number of epochs for different depths. Specifically, we show: (a) the Mean Absolute Error (MAE) loss for  $\alpha$  inference, (b) the Mean-Squared Logarithmic Error (MSLE) loss for  $K$  inference, and (c) the Weighted Cross-Entropy (WCE) loss for diffusion-state segmentation.



**Figure 5.** Loss values calculated on a test dataset consisting of 200 000 trajectories, as functions of the number of epochs for our Mamba-based and RNN-based models. Specifically, we show: (a) the Mean Absolute Error (MAE) loss for  $\alpha$  inference, (b) the mean-squared logarithmic error (MSLE) loss for  $K$  inference, and (c) the weighted cross-entropy (WCE) loss for diffusion-state segmentation.

logarithmic error (MSLE), and the segmentation loss is evaluated using the weighted cross-entropy error (WCE). The training is carried out on a subset of 200 000 trajectories generated using the AnDi-2 challenge simulation tools. The loss is calculated by summing the losses of the three blocks and back-propagated using the Adam optimizer. Segmentation is evaluated using the WCE loss, while the  $K$  regression is evaluated using the MSLE loss, and the  $\alpha$  regression is evaluated using the MAE loss. In figure 4, we evaluate the loss as a function of the depth  $N$  of the different blocks, from which we determine our best model to have a depth of  $N = 10$  for  $\alpha$  regression and  $N = 30$  for segmentation and  $K$  regression.

For fair benchmarking purposes, we compare our model (see figure 3) to a similar architecture but using bidirectional RNN instead of our Bi-Mamba block (see figure 2). As shown in figure 5, the Bi-Mamba model shows promising performances, scoring better overall in each category than a RNN-based model. Moreover, it shows its capabilities to train for a larger number of epochs without overfitting and with a smaller loss variance.

## 4. Conclusion

In this work, we implemented the structure and evaluated the performances of the Mamba architecture—a novel application of state-space models—towards the characterization of AnDi. Using a bidirectional scan mechanism, we demonstrated a notable efficiency in diffusion-state segmentation, as well as for effective-diffusion-coefficient and anomalous-exponent regression tasks. Improvement could be achieved through the use of specialized models, fine-tuned using simulations generated near the predictions of the generally-trained model presented in this work. Our contribution to the AnDi-2 challenge was designed in order to specifically evaluate the performances of Bi-Mamba models for AnDi. Our first model developed during the competition phase ranked 7th in  $\alpha$  inference (MAE of 0.27), 9th in  $K$  inference (MSLE of 0.05), 3rd in diffusion-type measurement [i.e. trapped with ( $\alpha < 0.2$ ), directed with  $\alpha > 1.8$ , or normal with  $\alpha = 1$  and with a F1 error of 0.91], and 10th on the change-point detection (root-mean-squared error of 2.7). Moreover, we expect improvements on the segmentation task by using Mamba-based Unet methods [34]. All together, this work contributes to the knowledge on, and continued development of, deep-learning methods for AnDi and physics in general.

## Data availability statement

The data that support the findings of this study are openly available at the following URL: <https://github.com/EMetBrown-Lab/Mamba-EMetBrown-ANDI2>.

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